

Finite Element Methods – APE3

Hilbert and Sobolev Spaces

Objectives of the session:

- Use the course material to prove existence of solutions via the Lax–Milgram theorem.
- Manipulate the concepts related to Hilbert spaces.
- Verify the assumptions of the Lax–Milgram theorem using results from the course or known algebraic results.
- Get a better overview of the Poincaré inequality

There exists a constant $C_P > 0$ (depending only on the dimension and Ω) such that

$$\|v\|_{L^2(\Omega)} \leq C_P \|\nabla v\|_{L^2(\Omega)} \quad \forall v \in H_0^1(\Omega).$$

1 Let us solve linear systems using Lax–Milgram

Exercise idea. Solve the linear system

$$Ku = f, \quad K \in \mathbb{R}^{n \times n}, \quad f \in \mathbb{R}^n, \quad u \in \mathbb{R}^n,$$

using a very (too!) powerful tool: the Lax–Milgram theorem. Clearly, the system admits a solution if and only if $\det K \neq 0$.

1. Recall why $\mathbb{R}^n$, equipped with the scalar product $(u | v)$, is a Hilbert space.

Solution. $\mathbb{R}^n$ with the standard scalar product

$$(u | v) = \sum_{i=1}^n u_i v_i$$

is an inner product space. Since $\mathbb{R}^n$ is finite-dimensional, it is complete for the induced norm $\|u\| = \sqrt{(u | u)}$. Alternatively you proved (back in the days in some Analysis course) that every Cauchy sequence in $\mathbb{R}^n$ do converge which is the definition of completeness. Therefore, $\mathbb{R}^n$ is a Hilbert space.

2. Define

$$a(u, v) = (Ku | v), \quad \ell(v) = (f | v),$$

and show that if a and ℓ satisfy the assumptions of the Lax–Milgram theorem, then there exists a unique $v \in \mathbb{R}^n$ solving the problem $Kv = f$.

Solution. If a is continuous and coercive on $\mathbb{R}^n$ and ℓ is continuous, the Lax–Milgram theorem guarantees the existence of a unique $v \in \mathbb{R}^n$ such that

$$a(v, w) = \ell(w) \quad \forall w \in \mathbb{R}^n.$$

In our case, this is equivalent to solving $Kv = f$, so uniqueness and existence are ensured under the Lax–Milgram hypotheses.

To go from the weak form to the linear system, one can choose $w = e_i \in \mathbb{R}^n$, a basis vector in which the matrix K is written; typically, the canonical basis is used.

3. (a) Show that $\ell(v)$ is bounded in the sense of the theorem. Which inequality do you use?

Solution. Using Cauchy–Schwarz:

$$|\ell(v)| = |(f | v)| \leq \|f\| \|v\|.$$

Hence ℓ is bounded with constant $\|f\|$.

solve

- (b) Show that $a(u, v)$ is continuous. Which inequality do you use?
(You will need $\|Ku\| \leq \|K\| \|u\|$. What is the meaning of $\|K\|$?)

Solution. We have

$$|a(u, v)| = |(Ku | v)| \leq \|Ku\| \|v\| \leq \|K\| \|u\| \|v\|.$$

Here, $\|K\|$ denotes the operator norm of K induced by the Euclidean norm: $\|K\| = \sup_{u \neq 0} \frac{\|Ku\|}{\|u\|}$. A remarkable result seen elsewhere assert that $\|K\|^2$ is the sum of the square of the coefficients of K . Therefore, a is continuous with constant $\|K\|$.

- (c) Assume that K is symmetric. Using the diagonalization theorem, show that K is coercive.

How does the coercivity constant β depend on the eigenvalues?

(You may use that $Q^T Q = \text{Id}$ for an orthogonal matrix Q .)

Solution. If K is symmetric, it can be diagonalized as $K = Q\Lambda Q^T$ with Q orthogonal and $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$. Then

$$a(u, u) = (Ku | u) = u^T Q \Lambda Q^T u = (Q^T u)^T \Lambda (Q^T u) = \sum_{i=1}^n \lambda_i (y_i)^2 \geq \min_i \lambda_i \|y\|^2,$$

with $y = Q^T u$. Since Q is orthogonal,

$$\|y\|^2 = (Q^T u | Q^T u) = u^T Q Q^T u = u^T \text{Id} u = (u | u) = \|u\|^2.$$

Hence a is coercive if all $\lambda_i > 0$, and the coercivity constant is $\beta = \min_i \lambda_i$. We deduce that

$$a(u, u) \geq \lambda_{\min}(K) \|u\|^2$$

- (d) Let K be a general square matrix. Define

$$K_{\text{sym}} = \frac{K^T + K}{2}.$$

Show that

$$a(u, u) = (Ku | u)$$

depends only on K_{sym} .

Solution. Let us write any square matrix K as

$$K = K_{\text{sym}} + K_{\text{antisym}}, \quad K_{\text{sym}} = \frac{K + K^T}{2}, \quad K_{\text{antisym}} = \frac{K - K^T}{2}.$$

Then for any $u \in \mathbb{R}^n$,

$$(Ku \mid u) = ((K_{\text{sym}} + K_{\text{antisym}})u \mid u) = (K_{\text{sym}}u \mid u) + (K_{\text{antisym}}u \mid u).$$

Since $K_{\text{antisym}}^T = -K_{\text{antisym}}$, we have

$$(u \mid K_{\text{antisym}}u) = u^T K_{\text{antisym}}u = -(K_{\text{antisym}}u)^T u = -u^T K_{\text{antisym}}u = 0.$$

Hence $(Ku \mid u) = (K_{\text{sym}}u \mid u)$, showing that $a(u, u)$ depends only on the symmetric part K_{sym} .

4. State a condition on the eigenvalues of K_{sym} ensuring that $a(\cdot, \cdot)$ is coercive.

Solution. a is coercive if and only if all eigenvalues of K_{sym} are strictly positive:

$$\lambda_{\min}(K_{\text{sym}}) > 0.$$

5. Is this condition equivalent to the nonvanishing of $\det K$, the product of the eigenvalues?

Solution. No. $\det K \neq 0$ only ensures that K is invertible, i.e., 0 is not an eigenvalue of K .

Coercivity requires *all eigenvalues of K_{sym} to be strictly positive*.

Example: $K = \text{diag}(1, -1)$ has $\det K = -1 \neq 0$, but $K_{\text{sym}} = K$ is not positive definite, so a is not coercive.

2 Let us solve $-\Delta u = f$ using Lax–Milgram

Exercise idea. Find a solution (*i.e.* show existence) to the boundary value problem below using a powerful functional analytic tool: the Lax–Milgram theorem. The idea is to rewrite the equation in weak form and interpret it as a variational problem on the Hilbert space $H_0^1(\Omega)$.

Let $\Omega \subset \mathbb{R}^n$ be a bounded Lipschitz domain. We consider the problem ($f \in L^2(\Omega)$)

$$\begin{cases} -\Delta u = f & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega. \end{cases}$$

The associated weak formulation is: find $u \in H_0^1(\Omega)$ such that

$$a(u, v) := \int_{\Omega} \nabla u \cdot \nabla v \, dx = \int_{\Omega} f v \, dx := \ell(v) \quad \forall v \in H_0^1(\Omega).$$

1. Recall the two definitions of $H_0^1(\Omega)$ and recall the inner product of this Hilbert space.

Explain why this space properly encodes the homogeneous Dirichlet boundary condition. Discuss why the weak formulation is elegant: the boundary condition is built into the functional framework, so it cannot be forgotten.

Solution.

There are two equivalent definitions of $H_0^1(\Omega)$:

Definition 1 (trace definition). $H_0^1(\Omega)$ is the subspace of $H^1(\Omega)$ formed by functions whose trace on the boundary vanishes:

$$H_0^1(\Omega) = \{u \in H^1(\Omega) : u|_{\partial\Omega} = 0\}.$$

For Lipschitz domains, these two definitions are equivalent.

Definition A (closure definition). $H_0^1(\Omega)$ is the closure of $C_c^1(\Omega)$ in $H^1(\Omega)$:

$$H_0^1(\Omega) = \overline{C_c^1(\Omega)}^{H^1}.$$

In other words, it consists of all H^1 functions that can be approximated (in the H^1 norm) by smooth functions with compact support inside Ω .

Hilbert structure. The space $H_0^1(\Omega)$ is a Hilbert space with inner product

$$(u, v)_{H_0^1(\Omega)} = \int_{\Omega} \nabla u \cdot \nabla v \, dx.$$

Since $H_0^1(\Omega)$ is a closed subspace of the Hilbert space $H^1(\Omega)$, it is itself a Hilbert space with the same inner product. Completeness was mentioned in class but never proven.

Moreover, the norm

$$\|u\|_{H_0^1} := \|\nabla u\|_{L^2(\Omega)}$$

is equivalent to the full H^1 norm on $H_0^1(\Omega)$; we won't need that fact.

Encoding of the boundary condition.

The homogeneous Dirichlet condition $u = 0$ on $\partial\Omega$ is built directly into the space $H_0^1(\Omega)$. Either by the trace or by the fact that a dense class $C_c^1(\Omega)$ vanishes on the boundary. This means that when we look for a weak solution $u \in H_0^1(\Omega)$, the boundary condition is automatically satisfied.

This is one of the elegant aspects of the weak formulation: the admissible space already incorporates the boundary condition. As a consequence, we do not need to impose $u = 0$ separately in the variational equation — it is built into the functional framework and therefore cannot be forgotten.

2. Define

$$a(u, v) = \int_{\Omega} \nabla u \cdot \nabla v \, dx, \quad \ell(v) = \int_{\Omega} f v \, dx,$$

and show that if a and ℓ satisfy the assumptions of the Lax–Milgram theorem, then there exists a unique $u \in H_0^1(\Omega)$ solving the weak problem.

Does this weak solution solve the PDE in the classical sense?

Solution.

We consider the bilinear form

$$a(u, v) = \int_{\Omega} \nabla u \cdot \nabla v \, dx, \quad \ell(v) = \int_{\Omega} f v \, dx,$$

defined on $H_0^1(\Omega)$.

If a is continuous and coercive on $H_0^1(\Omega)$ and if ℓ is continuous, then the Lax–Milgram theorem ensures the existence of a unique

$$u \in H_0^1(\Omega)$$

such that

$$a(u, v) = \ell(v) \quad \forall v \in H_0^1(\Omega).$$

In other words,

$$\int_{\Omega} \nabla u \cdot \nabla v \, dx = \int_{\Omega} f v \, dx \quad \forall v \in H_0^1(\Omega),$$

which is exactly the weak formulation of the Poisson problem with homogeneous Dirichlet boundary condition. Therefore, under the assumptions of the Lax–Milgram theorem, there exists a unique weak solution.

Does the weak solution solve the PDE in the classical sense?

Not necessarily. The weak solution u belongs to $H_0^1(\Omega)$, which does not guarantee that second derivatives exist in the classical sense. Therefore, $-\Delta u = f$ may only hold in the weak (distributional) sense.

However, if f and the domain Ω are sufficiently regular (for instance $f \in L^2(\Omega)$ and Ω is smooth enough ; this follows from the so called elliptic estimates), additional elliptic regularity results imply that $u \in H^2(\Omega)$. In that case, the weak formulation

$$\int_{\Omega} \nabla u \cdot \nabla v \, dx = \int_{\Omega} f v \, dx \quad \forall v \in H_0^1(\Omega)$$

can be rewritten, after integration by parts (which is justified when $u \in H^2(\Omega)$), as

$$\int_{\Omega} (-\Delta u - f) v \, dx = 0 \quad \forall v \in C_c^1(\Omega).$$

By the L^2 version of the fundamental lemma of the calculus of variations seen during CM1, this implies

$$-\Delta u = f \quad \text{almost everywhere in } \Omega.$$

Hence u is a *strong* solution, meaning that the PDE holds almost everywhere.

If the solution is further known to be of class $C^2(\Omega)$, then the Laplacian is defined pointwise and the PDE holds in the classical sense.

Example (right-hand side with a jump).

Let $\Omega = (-1, 1)$ and let f be the Heaviside function

$$f(x) = \begin{cases} 0 & \text{if } x < 0, \\ 1 & \text{if } x > 0. \end{cases}$$

We consider

$$-u'' = f \quad \text{in } (-1, 1), \quad u(-1) = u(1) = 0.$$

Since $f \in L^2(-1, 1)$, the Lax–Milgram theorem ensures the existence of a unique weak solution $u \in H_0^1(-1, 1)$.

We can solve explicitly on each subinterval:

- On $(-1, 0)$: $u'' = 0$, so u is affine. - On $(0, 1)$: $u'' = -1$, so u is a quadratic polynomial. Imposing continuity of u and u' at $x = 0$, and the boundary conditions at ± 1 , determines a unique solution. One checks that:

- u is C^1 on $(-1, 1)$, - u'' exists and equals 0 on $(-1, 0)$, - u'' exists and equals -1 on $(0, 1)$. However, u'' has a jump at $x = 0$, so u is *not* $C^2(-1, 1)$.

Therefore, the equation $-u'' = f$ does not hold in the classical (pointwise C^2) sense on the whole interval. It holds only almost everywhere.

This example shows that even for a simple L^2 right-hand side, the weak solution may fail to be twice continuously differentiable, which explains why the weak formulation is the natural framework.

3. Show that $\ell(v)$ is bounded in the sense of the theorem. Which inequalities do you use?

Solution. We have

$$\ell(v) = \int_{\Omega} f v \, dx.$$

Using the Cauchy–Schwarz inequality in $L^2(\Omega)$,

$$|\ell(v)| \leq \|f\|_{L^2(\Omega)} \|v\|_{L^2(\Omega)}.$$

Since $v \in H_0^1(\Omega)$, the Poincaré inequality gives

$$\|v\|_{L^2(\Omega)} \leq C_P \|\nabla v\|_{L^2(\Omega)}.$$

Therefore,

$$|\ell(v)| \leq C_P \|f\|_{L^2(\Omega)} \|\nabla v\|_{L^2(\Omega)}.$$

Hence ℓ is continuous (bounded) on $H_0^1(\Omega)$.

The inequalities used are:

- Cauchy–Schwarz inequality,
- Poincaré inequality.

Did you know?

The best (smallest) constant C_P in the Poincaré inequality

$$\|v\|_{L^2(\Omega)} \leq C_P \|\nabla v\|_{L^2(\Omega)}, \quad \forall v \in H_0^1(\Omega),$$

is related to the smallest positive eigenvalue λ_1 of the Laplacian with homogeneous Dirichlet boundary conditions:

$$-\Delta \phi = \lambda \phi \quad \text{in } \Omega, \quad \phi = 0 \quad \text{on } \partial\Omega.$$

In fact, one can show that

$$C_P = \frac{1}{\sqrt{\lambda_1}}.$$

This is a nice example where spectral theory also appears naturally in the non-finite dimensional analysis: the Poincaré constant is determined by the “lowest frequency” of the domain.

The link between the above eigenvalue problem and the optimal constant follows by considering the constrained variational problem studied during the CM1

$$C'_P = \inf \{ \|\nabla v\|_{L^2(\Omega)} : v \in H_0^1(\Omega) \, \|v\|_{L^2(\Omega)} = 1 \};$$

the eigenvalue λ is exactly the Lagrange multiplier ! Note that in general when one writes a proof of the Poincaré inequality one finds a constant C_P which satisfies $C_P \geq C'_P$; it is then a nontrivial task to assert that $C_P = C'_P$.

4. Show that $a(u, v)$ is continuous. Which inequality do you use?

Solution. We have

$$a(u, v) = \int_{\Omega} \nabla u \cdot \nabla v \, dx.$$

Applying the Cauchy–Schwarz inequality in $L^2(\Omega)^n$,

$$|a(u, v)| \leq \|\nabla u\|_{L^2(\Omega)} \|\nabla v\|_{L^2(\Omega)}.$$

Thus,

$$|a(u, v)| \leq \|u\|_{H_0^1(\Omega)} \|v\|_{H_0^1(\Omega)}.$$

Hence a is continuous on $H_0^1(\Omega)$.

The inequality used is the Cauchy–Schwarz inequality.

5. Show that $a(u, v)$ is coercive. Which inequality do you use?

Solution. For $u \in H_0^1(\Omega)$,

$$a(u, u) = \int_{\Omega} |\nabla u|^2 \, dx = \|\nabla u\|_{L^2(\Omega)}^2.$$

Therefore,

$$a(u, u) = \|\nabla u\|_{L^2(\Omega)}^2 = \|u\|_{H_0^1(\Omega)}^2 \geq C^2 \|u\|_{H_0^1(\Omega)}^2$$

for some constant $C > 0$ ($C = 1$ is a reasonable choice but any $C \in (0, 1]$ is ok).

Hence a is coercive on $H_0^1(\Omega)$.

No inequalities were used.

6. Let V be a finite-dimensional subspace of $H_0^1(\Omega)$; this is the idea behind the Galerkin method and the finite element method. Does there exist a unique solution $u \in V$ to

$$\int_{\Omega} \nabla u \cdot \nabla v \, dx = \int_{\Omega} f v \, dx \quad \forall v \in V?$$

Solution. Since V is a finite-dimensional subspace of the Hilbert space $H_0^1(\Omega)$, it is itself a Hilbert space (any finite-dimensional normed space is complete).

Existence and uniqueness.

Define the bilinear form

$$a(u, v) = \int_{\Omega} \nabla u \cdot \nabla v \, dx, \quad u, v \in V,$$

and the linear form

$$\ell(v) = \int_{\Omega} f v \, dx, \quad v \in V.$$

- a is continuous on V (Cauchy–Schwarz inequality). - a is coercive on V : for all $v \in V$,

$$a(v, v) = \int_{\Omega} |\nabla v|^2 \, dx \geq 0,$$

and equality holds only if $v = 0$ (since $V \subset H_0^1(\Omega)$ and the H^1 norm is a norm on H_0^1).

Since V is finite-dimensional, it is automatically complete. Therefore, by the Lax–Milgram theorem, there exists a unique $u \in V$ such that

$$a(u, v) = \ell(v) \quad \forall v \in V.$$

Remark on Cauchy sequences.

Any Cauchy sequence $(u_n)_n \subset V$ with respect to the H_0^1 norm converges in $H_0^1(\Omega)$ because H_0^1 is complete. But since V is finite-dimensional, the limit must belong to V ; write the sequence in coordinate in any basis of V . This proves that V is complete, justifying the application of Lax–Milgram in finite-dimensional subspaces (as in the FEM setting).

3 Let us solve $-\Delta u + Vu = f$ using Lax–Milgram

Exercise idea. Show existence to the system below by *perturbing* the arguments from the previous section.

Let $\Omega \subset \mathbb{R}^n$ be a bounded Lipschitz domain, and let $V \in C^0(\bar{\Omega})$ be a given potential. We consider the problem ($f \in L^2(\Omega)$)

$$\begin{cases} -\Delta u + Vu = f & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega. \end{cases}$$

The associated weak formulation is: find $u \in H_0^1(\Omega)$ such that

$$a(u, v) := \int_{\Omega} \nabla u \cdot \nabla v \, dx + \int_{\Omega} Vuv \, dx = \int_{\Omega} f v \, dx = \ell(v) \quad \forall v \in H_0^1(\Omega).$$

1. Recall that $\ell(v)$ is bounded in the sense of the theorem by using the result of the previous section.
2. Show that $a(u, v)$ is continuous.

Solution. We have

$$a(u, v) = \int_{\Omega} \nabla u \cdot \nabla v \, dx + \int_{\Omega} Vuv \, dx.$$

- For the first term, the Cauchy–Schwarz inequality gives

$$\left| \int_{\Omega} \nabla u \cdot \nabla v \, dx \right| \leq \|\nabla u\|_{L^2(\Omega)} \|\nabla v\|_{L^2(\Omega)}.$$

- For the second term, using Cauchy–Schwarz in $L^2(\Omega)$ and the fact that $V \in C^0(\bar{\Omega}) \subset L^\infty(\Omega)$,

$$\left| \int_{\Omega} Vuv \, dx \right| \leq \|V\|_{L^\infty(\Omega)} \|u\|_{L^2(\Omega)} \|v\|_{L^2(\Omega)} \leq \|V\|_{L^\infty(\Omega)} C_P^2 \|\nabla u\|_{L^2(\Omega)} \|\nabla v\|_{L^2(\Omega)},$$

where we used the Poincaré inequality $\|u\|_{L^2} \leq C_P \|\nabla u\|_{L^2}$.

Combining both terms, we get

$$|a(u, v)| \leq (1 + C_P^2 \|V\|_{L^\infty(\Omega)}) \|\nabla u\|_{L^2(\Omega)} \|\nabla v\|_{L^2(\Omega)}.$$

Thus a is continuous on $H_0^1(\Omega)$.

3. Show that if $V \geq 0$ for all $u \in H_0^1(\Omega)$, $a(u, u)$ is coercive.

Solution. For $u \in H_0^1(\Omega)$ we compute

$$a(u, u) = \int_{\Omega} |\nabla u|^2 dx + \int_{\Omega} V u^2 dx.$$

If $V \geq 0$ almost everywhere in Ω , then the second term is non-negative, hence

$$a(u, u) \geq \int_{\Omega} |\nabla u|^2 dx = \|\nabla u\|_{L^2(\Omega)}^2.$$

One concludes as in the previous section.

4. Show that for all $u \in H_0^1(\Omega)$, $a(u, u)$ satisfies $a(u, u) \geq (1 - C_P^2 \|V_-\|_{\infty}) \|\nabla u\|_{L^2(\Omega)}^2$, and state a condition on V for a to be coercive. *The condition should be weaker than $V \geq 0$.*

Solution. Recall the notations:

$$\|u\|_{\infty} := \sup_{x \in \Omega} |u(x)|, \quad \mathcal{V}_- := \inf_{x \in \Omega} V(x) \leq 0, \quad V_-(x) := \max(0, -V(x)) = -\mathcal{V}_-.$$

We have

$$a(u, u) = \int_{\Omega} |\nabla u|^2 dx + \int_{\Omega} V u^2 dx.$$

Since $V_-(x) \geq 0$ is the negative part of V , we can bound

$$\int_{\Omega} V u^2 dx \geq - \int_{\Omega} V_- u^2 dx \geq -\|V_-\|_{\infty} \int_{\Omega} |u|^2 dx \leq -\|V_-\|_{\infty} C_P^2 \|\nabla u\|_{L^2(\Omega)}^2,$$

where we used the Poincaré inequality $\|u\|_{L^2(\Omega)} \leq C_P \|\nabla u\|_{L^2(\Omega)}$.

Therefore, for all $u \in H_0^1(\Omega)$,

$$a(u, u) \geq \|\nabla u\|_{L^2(\Omega)}^2 - \|V_-\|_{\infty} C_P^2 \|\nabla u\|_{L^2(\Omega)}^2 = (1 - C_P^2 \|V_-\|_{\infty}) \|\nabla u\|_{L^2(\Omega)}^2.$$

Condition for coercivity: The bilinear form a is coercive if

$$\|V_-\|_{\infty} < \frac{1}{C_P^2} \quad \text{or equivalently} \quad -\mathcal{V}_- < \frac{1}{C_P^2}.$$

We can define the coercivity constant

$$\beta := 1 - C_P^2 \|V_-\|_{\infty} > 0,$$

so that

$$a(u, u) \geq \beta \|\nabla u\|_{L^2(\Omega)}^2 \quad \forall u \in H_0^1(\Omega).$$

Remark: If $V \geq 0$ everywhere, then $\mathcal{V}_- = 0$, $\|V_-\|_{\infty} = 0$, and $\beta = 1$. Coercivity is immediate.

Did you know?

The best Poincaré constant is linked to the first eigenvalue of the Laplacian. This means that, in order for our physical problem to have a solution, the potential V must satisfy a *spectral condition* (as we encountered in the first exercise!)

- Are there deep connections between spectral geometry and quantum physics? Absolutely! Eigenvalues of the Laplacian are central in quantum mechanics.
- Is this part of our FEM course? Not really—but it shows how math and physics are connected.

Think of it this way: the geometry of the domain and the potential together “control” the solution, via their spectrum.

5. Using your condition on V state that there exists a unique solution to the weak form of the problem using the Lax-Milgram theorem.